Unit - 5 : HYPOTHESIS TESTING

Session - 10

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Post-Hoc Tests

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Why Post-Hoc Test?

ANOVA tells you only one thing:

“At least one group is different.”

But ANOVA *does not* tell you:

- Which group?
- How many groups?
- Which specific pairs differ?

It's like saying:

“Something is wrong in the class.”
But not telling **which student** or **what exactly is wrong**.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Post-hoc test

- A post hoc is conducted after the null hypothesis of ANOVA is rejected to determine the different treatments(s)

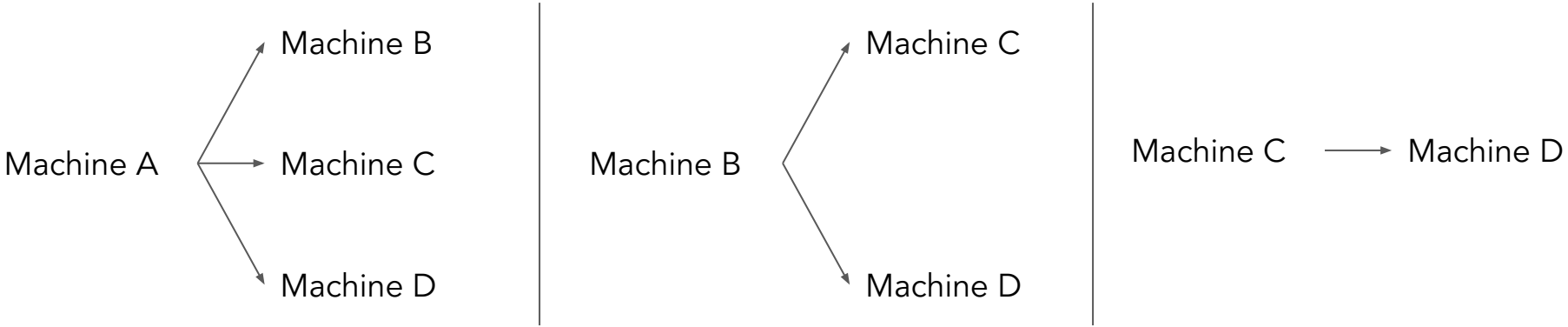
- There are various post hoc tests available such as:
  - Tukey's HSD test (Tukey's Honest(ly) Significant Difference test)
  - Scheffe test
  - Duncan's Multiple Range test
  - Fisher's' LSD test (Fisher's Least Significant Difference test)
  - Bonferroni test

- We will study the Tukey's HSD test in detail

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Post-hoc test

- Consider our example where Ryan wants to find out the which machines had different result
- Each pair of machines is tested for the statistical difference



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Post-hoc test

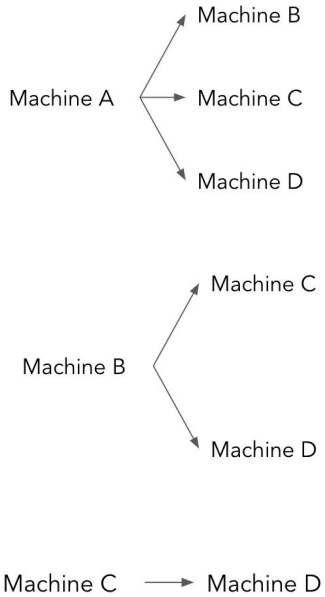
Thus the test hypothesis are

$H_{01}: \mu_{machine_A} = \mu_{machine_B}$ Against $H_{11}: \mu_{machine_A} \neq \mu_{machine_B}$

$H_{02}: \mu_{machine_A} = \mu_{machine_C}$ Against $H_{12}: \mu_{machine_A} \neq \mu_{machine_C}$

$H_{03}: \mu_{machine_A} = \mu_{machine_D}$ Against $H_{13}: \mu_{machine_A} \neq \mu_{machine_D}$

$H_{04}: \mu_{machine_B} = \mu_{machine_C}$ Against $H_{14}: \mu_{machine_A} \neq \mu_{machine_C}$



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Post-hoc test

The test statistic is:

Obtained from the Tukey table

$$T_\alpha = q_{\alpha,(t,f)} \sqrt{\frac{MSE}{n}}$$

t: total treatments

f: error degrees of freedom

MSE: Mean error sum of squares (from ANOVA table)

n: number of observations in a group

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Post-hoc test

- Consider the absolute difference between two treatments $|\bar{x}_i - \bar{x}_j|$
- The decision rule: Reject H_0, if the absolute difference $\geq T_\alpha$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Post-hoc test

The output is as follows:

```
    Multiple Comparison of Means - Tukey HSD, FWER=0.05
=============================================================
  group1     group2  meandiff p-adj   lower    upper   reject
-------------------------------------------------------------
machine_A machine_B    -11.1 0.0044 -18.8842  -3.3158   True
machine_A machine_C   -18.74  0.001 -26.5242 -10.9558   True
machine_A machine_D     5.46 0.2265  -2.3242  13.2442  False
machine_B machine_C    -7.64 0.0553 -15.4242   0.1442  False
machine_B machine_D    16.56  0.001   8.7758  24.3442   True
machine_C machine_D     24.2  0.001  16.4158  31.9842   True
-------------------------------------------------------------
```

True: reject H_0

False: fail to reject H_0 (accept H_0)

It can been seen that there is statistical difference between pairs of machines (A,B), (A,C), (B,D), and (C,D).

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Example

CLUB 1	CLUB 2	CLUB 3
254	234	200
263	218	222
241	235	197
237	227	206
251	216	204

$X_1 = 249.2 \quad n_1 = 5$

$X_2 = 226.0 \quad n_2 = 5$

$X_3 = 205.8 \quad n_3 = 5$

$n = 15$

$\bar{\bar{X}} = 227.0 \quad c = 3$

$$SSB = 5\,(249.2 - 227)^2 + 5\,(226 - 227)^2 + 5\,(205.8 - 227)^2 = 4716.4$$

$$SSW = (254 - 249.2)^2 + (263 - 249.2)^2 + \cdots + (204 - 205.8)^2 = 1119.6$$

MSB = 4716.4 / (3-1) = 2358.2

MSW = 1119.6 / (15-3) = 93.3

$$F_{STAT} = \frac{2358.2}{93.3} = 25.275$$

Decision:

Reject H_0 at $\alpha = 0.05$

Conclusion:

There is evidence that at least one μ_j differs from the rest

Find:

Which are the groups differ?

This file is meant for personal use by dominic3097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Solution:

We proceed with Tukey's Test to find the differed group

$$T_\alpha = q_{\alpha,(t,f)}\sqrt{\frac{MSE}{n}}$$

Treatments t=3

df_{within} = 12

α =0.05

q=3.77

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

*Table

MSW = 93.3

n= 5

$T_\alpha = 3.77 \times \sqrt{(93.3/5)}$

$T_\alpha = 3.77 \times 4.320 = 16.29$

Note: if the difference between two group means > **16.29**, that pair is significantly different at α = 0.05.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Compute absolute differences between means:

- **Club1 − Club2** = |249.2 − 226.0| = **23.2**
- **Club1 − Club3** = |249.2 − 205.8| = **43.4**
- **Club2 − Club3** = |226.0 − 205.8| = **20.2**

Compare each to HSD = **16.29**:

- 23.2 > 16.29 → **Significant** (Club1 vs Club2)
- 43.4 > 16.29 → **Significant** (Club1 vs Club3)
- 20.2 > 16.29 → **Significant** (Club2 vs Club3)

Conclusion:

ANOVA showed a statistically significant difference among the three clubs.

Tukey's post-hoc test shows **all three pairwise comparisons are significant**.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

PLEASE!! NOTE

- For equal number of observations in each treatment, tukey HSD test can be used
- However when the data is unequal it is not efficient
- In such a scenario, one may use the Scheffe test

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Extra - Post Read

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Kruskal Wallis H test

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Kruskal Wallis H test

- It is the non-parametric equivalent of ANOVA
- Used when two or more independent samples when population is non-normal
- The test hypothesis remains the same as that of ANOVA

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Kruskal Wallis H test - Procedure

- Let n_i (i = 1, 2, … ,t) denote the number of observations a treatment
- Combine all t samples so that there are $n = n_1 + n_2 + \ldots + n_t$ units in the sample
- Arrange it in ascending order and rank them (rank 1 is given to the smallest observation, solve for ties)
- The test statistic is given by

sum of ranks of observations in the ith sample

$$H = \frac{12}{n(n+1)} \sum \frac{R_i^2}{n_i} - 3(n+1)$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Kruskal Wallis H test

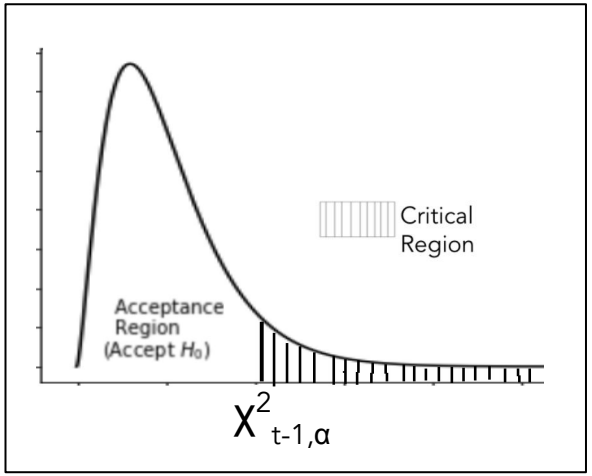
- Note that each $n_i > 5$

- Decision rule: Reject H_0 if $H > \chi^2_{t-1,\alpha}$

  or

  Reject H_0 if p-value $\leq \alpha$

Where, α is the level of significance (l.o.s.)



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

PLEASE!! NOTE

The python code to conduct kruskal wallis H test is

scipy.stats.kruskal(sample$_1$, sample$_2$,..., sample$_n$)

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Kruskal wallis H test

Question:

Ryan is a production manager at an industry manufacturing alloy seals. They have 3 machines - A, B and C. Ryan wants to study whether all the machines have equal efficiency. There no information about the tensile strength (in N/m^2) of the population. Can it be said that the machines are produce the sample tensile strength?

A	B	C
72.5	33.2	24.5
84.8	23.4	34.6
76.6	35.7	
34.7	33.7	
	34.5	

Test at 5% level of significance.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Kruskal wallis H test

Solution:

Ryan is a production manager at an industry manufacturing alloy seals. They have 4 machines - A, B, and C

Let μ_1 be the average tensile strength due to machine A
μ_2 be the average tensile strength due to machine B
μ_3 be the average tensile strength due to machine C

To test, H_0: $\mu_1 = \mu_2 = \mu_3$ Against H_1: At least one μ_i is different (i=1,2,3)

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Kruskal wallis H test

Solution:

First combine the sample and rank it.

Here $n_1 = 4$, $n_2 = 5$, $n_3 = 2$

Thus $n = 4+5+2 = 11$

$R_1 = 7+9+10+11 = 37$

$R_2 = 1+3+4+5+8 = 21$

$R_3 = 2+6 = 8$

Sample treatment	Observation	Rank
B	23.4	1
C	24.5	2
B	33.2	3
B	33.7	4
B	34.5	5
C	34.6	6
A	34.7	7
B	35.7	8
A	72.5	9
A	76.6	10
A	84.8	11

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Kruskal wallis H test

Solution:

The test statistic is $$H = \frac{12}{n(n+1)} \sum \frac{R_i^2}{n_i} - 3(n+1)$$

$$= \frac{12}{11(11+1)} \left[\frac{R_1^2}{n_1} + \frac{R_2^2}{n_2} + \frac{R_3^2}{n_3} \right] - 3(11+1)$$

$$= \frac{12}{11(11+1)} \left[\frac{37^2}{4} + \frac{21^2}{5} + \frac{8^2}{2} \right] - 3(11+1)$$

$$= 6.041$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Kruskal wallis H test

Solution:

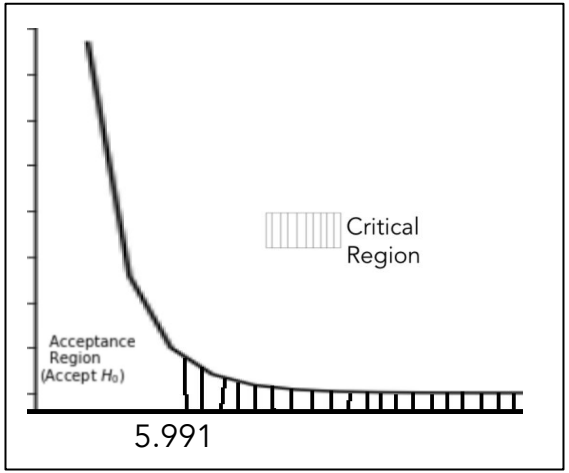
The test statistic H = 6.041

From the chi-square table we have $\chi^2_{t-1,\alpha} = \chi^2_{3-1,0.05} = 5.991$

Since $H > \chi^2_{t-1,\alpha}$ (6.041 > 5.991), we fail to reject H_0

The associate p-value is 0.048
Since p-value < 0.05, we reject H_0.

Thus there is enough evidence to conclude that the tensile strength due to machines is different



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

PLEASE!! NOTE

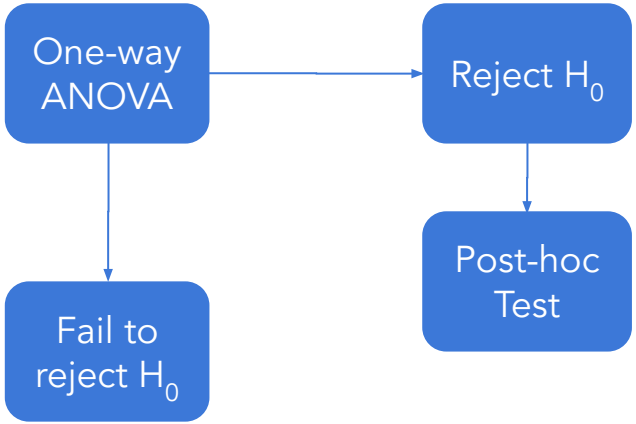
Post-hoc tests

	Parametric test	Non-parametric test
Equal groups	Tukey HSD	Conover test
Unequal groups	Scheffe test	Dunn's multiple Comparison test

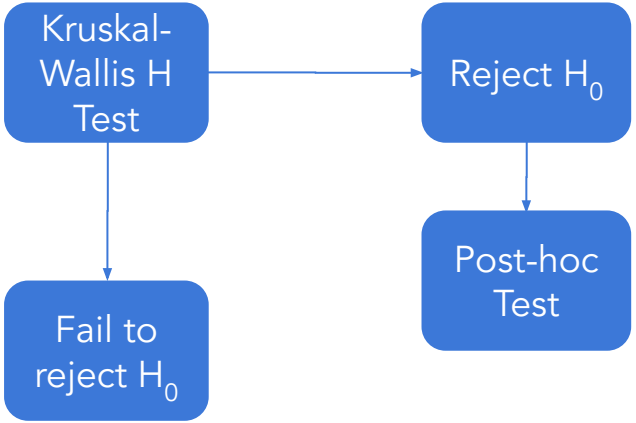
This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Summary

Parametric Test



Non-parametric Test



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

One - way ANOVA

- Ryan only considered the effect of the machines on the tensile strength
- What if he considers the effect of work shifts used?
- The effect of the quality of material and the effect due to machine can be tested using two way ANOVA

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Factorial Design: Two-Way ANOVA

Examines the effect of

- Two factors of interest on the dependent variable
  - e.g., Percent carbonation and line speed on soft drink bottling process
- Interaction between the different levels of these two factors
  - e.g., Does the effect of one particular carbonation level depend on which level the line speed is set?

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Test of Hypothesis and Assumptions: Two-Way ANOVA

- Just as we had Sums of Squares and Mean Squares in One-Way ANOVA, we have the same in Two-Way ANOVA
- Recall, Mean Squares are measures of variability across the levels of the relevant factor of interest
- In Two-Way ANOVA, we measure the overall variability in the data

Assumptions

- Populations are normally distributed
- Populations have equal variances
- Independent random samples are drawn

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two-Way ANOVA: Sources of Variation

Two Factors of interest: A and B

r = number of levels of factor A

c = number of levels of factor B

n′ = number of replications for each cell

n = total number of observations in all cells $n = (r)(c)(n')$

X_{ijk} = value of the kth observation of level i of factor A and level j of factor B

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two-Way ANOVA: Sources of Variation

$$SST = SSA + SSB + SSAB + SSE$$

SST
Total Variation

n - 1

SSA
Factor A Variation

SSB
Factor B Variation

SSAB
Variation due to interaction between A and B

SSEB
Random Variation (Error)

Degrees of Freedom:

r – 1

c -1

(r – 1) (c – 1)

rc (n' – 1)

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Mean Square Calculations

$$\text{MSA} = \text{Mean square factor A} = \frac{\text{SSA}}{r-1}$$

$$\text{MSB} = \text{Mean square factor B} = \frac{\text{SSB}}{c-1}$$

$$\text{MSAB} = \text{Mean square interaction} = \frac{\text{SSAB}}{(r-1)(c-1)}$$

$$\text{MSE} = \text{Mean square error} = \frac{\text{SSE}}{rc(n'\ 1)}$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two-Way ANOVA: The F Test Statistics

F Test for Factor A Effect

$H_0: \mu_{1..} = \mu_{2..} = \mu_{3..} = \cdots = \mu_{r..}$

H_1: Not all $\mu_{i..}$ are equal

$$F_{STAT} = \frac{MSA}{MSE}$$

Reject H_0 if $F_{STAT} > F_\alpha$

F Test for Factor B Effect

$H_0: \mu_{.1.} = \mu_{.2.} = \mu_{.3.} = \cdots = \mu_{.c.}$

H_1: Not all $\mu_{.j.}$ are equal

$$F_{STAT} = \frac{MSB}{MSE}$$

Reject H_0 if $F_{STAT} > F_\alpha$

F Test for Interaction Effect

H_0: the interaction of A and B Is equal to zero

H_1: interaction of A and B is not zero

$$F_{STAT} = \frac{MSAB}{MSE}$$

Reject H_0 if $F_{STAT} > F_\alpha$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two-Way ANOVA Summary Table

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F
Factor A	SSA	r – 1	MSA = SSA /(r – 1)	$\frac{MSA}{MSE}$
Factor B	SSB	c – 1	MSB = SSB /(c – 1)	$\frac{MSB}{MSE}$
AB (Interaction)	SSAB	(r – 1)(c – 1)	MSAB = SSAB / (r – 1)(c – 1)	$\frac{MSAB}{MSE}$
Error	SSE	rc(n′ – 1)	MSE = SSE/rc(n’ – 1)	
Total	SST	n – 1		

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

The business is thriving, and Ryan finds it crucial that the production should progress without any breaks. He decides to have three shifts - Morning, Afternoon and Night shift.

Now there are two factors that may affect the tensile strength of the alloy wire.

- The production machine
- The workforce at each shift

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

- In two way ANOVA the effect due to two nominal variables can be tested over a numeric variable
- Ryan can test for effect of machine on the tensile strength while he tests for effect due to shifts
- Now the effect of machines can be considered as the treatments and the effect due to shifts are the blocks

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Blocking

- Previously, Ryan had only one shift that was working. So there was no possible variation that could occur
- Now when he has 3 shifts the and possible variation due to these shifts may arise
- The data becomes heterogeneous and can be made homogeneous
- Blocking is used when the data is not homogeneous

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

- One of the null hypothesis to be tested is

H_{01}: The averages of all treatments are same.
i.e. $\alpha_1 = \alpha_2 = \ldots = \alpha_n$

Against

H_{11}: At the least one treatment has a different average

- Failing to reject H_{01}, implies that all treatment have the same average

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

- Another null hypothesis to be tested is

H_{02}: The averages of all blocks are same.
i.e. $\beta_1 = \beta_2 = \ldots = \beta_n$

Against

H_{12}: At the least one block has a different average

- Failing to reject H_{02}, implies that all blocks have the same average

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

- Suppose Ryan collects data for tensile strength of wires produced by each machine at each shift

- There are 4 treatments (t = 4) (between M/C)

- There are 3 blocks (b = 3) (between Shift)

	A	B	C	D
Morning Shift	68.7	78.7	55.9	84.7
Afternoon Shift	75.4	75.4	56.1	85.3
Night Shift	80.9	80.9	57.3	87.9

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

- Each treatment has 5 observations (n_i = 5) where i = 1, 2, …, t
- Each block as 4 observations (b_j = 4) where j = 1, 2, …, b
- Total number of observations is given by N

$$N = \sum_i^t \sum_j^b n_{ij}$$

	A	B	C	D
Morning Shift	68.7	78.7	55.9	84.7
Afternoon Shift	75.4	75.4	56.1	85.3
Night Shift	80.9	80.9	57.3	87.9

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

- Let α_i (i=1, 2, …, t) denote the average strength due to each machine and β_j dnote the average strength due to the shift
- For our example, t = 4 and b = 3
- The test hypothesis can be written as

	A	B	C	D
Morning Shift	68.7	78.7	55.9	84.7
Afternoon Shift	75.4	75.4	56.1	85.3
Night Shift	80.9	80.9	57.3	87.9

$H_{01}: \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4$ Against H_{11}: At least α_i is different

$H_{02}: \beta_1 = \beta_2 = \beta_3$ Against H_{12}: At least β_j is different

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

- In one way ANOVA, the entire population variance is split into two component
  - Variation due to treatments
  - Variation due to blocks
  - Error variation

- Total variation = Treatment Variation + Block Variation + Error Variation

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Total variation

- It is the total sum of squares (TSS)
- Let x_{ij} be the observations in the ith treatment and j^{th} block
- $\bar{x}_{..}$ is the grand mean, i.e. the mean of all observations
- The total variation is given by

$$TSS = \sum_i^t \sum_j^b (x_{ij} - \bar{x}_{..})^2$$

Summation over the treatments

Summation over the blocks

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Treatment variation

- It is the treatment sum of squares (TrSS)
- Let $x_{i.}$ be the observations in the ith treatment with b observations in each treatment and $\bar{x}_{i.}$ is the mean over i^{th} treatment
- $\bar{x}_{..}$ is the grand mean, i.e. the mean of all observations
- The treatment variation is given by

$$TrSS = \sum_i^t b(\bar{x}_{i.} - \bar{x}_{..})^2$$

Summation over all treatments

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Block variation

- It is the block sum of squares (BSS)
- Let $x_{.j}$ be the observations in the jth block with t observations in each block and $\bar{x}_{.j}$ is the mean over j^{th} block
- $\bar{x}_{..}$ is the grand mean, i.e. the mean of all observations
- The block variation is given by

$$BSS = \sum_j^b t(\bar{x}_{.j} - \bar{x}_{..})^2$$

Summation over all blocks

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Error variation

- It is the error sum of squares (ESS)
- Let $x_{i.}$ be the observations in the ith treatment and $\bar{x}_{.j}$ is the mean over j^{th} block
- $\bar{x}_{..}$ is the grand mean, i.e. the mean of all observations
- The error sum of squares is given by

$$ESS = \sum_i^t \sum_j^b (x_{ij} - \bar{x}_{i.} - \bar{x}_{.j} + \bar{x}_{..})^2$$

Summation over all treatments

Summation over all blocks

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

PLEASE!! NOTE

Error sum of squares

During problem solving, the error sum of squares is obtained as:

$$\mathrm{ESS} = \mathrm{TSS} - \mathrm{TrSS} - \mathrm{BSS}$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

- The test statistic for treatments is given by

$$\text{F-ratio} = \frac{\frac{TrSS}{df_{Tr}}}{\frac{ESS}{df_e}} = \frac{\text{MTrSS}}{\text{MESS}}$$

Mean Treatment Sum of Squares (s^2_t)

Mean Error Sum of Squares (s^2_e)

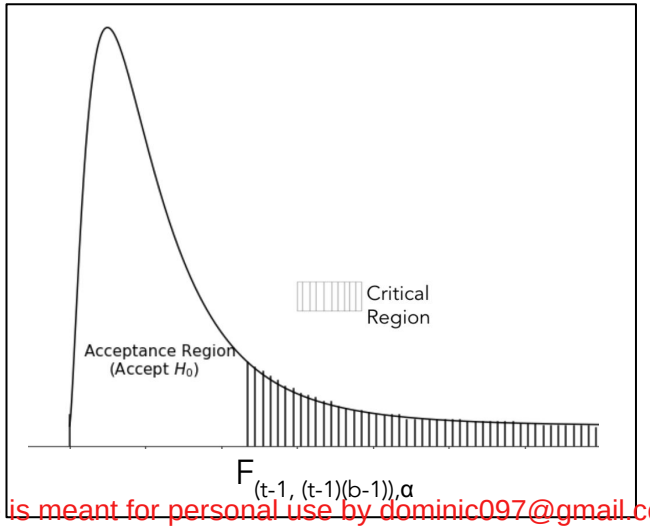
- Under H_0, the test statistic follows F-distribution with (df_{Tr}, df_e) degrees of freedom

- Decision Rule: If $F_{cal} \geq F_{(t-1,\ (t-1)(b-1))\alpha}$ then we reject H_0 at α% level of significance

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

Decision Rule: If $F_{cal} \geq F_{(t-1,\ (t-1)(b-1)),\alpha}$ or p-value ≤ α, then we reject H_0 at α% level of significance.



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

- The test statistic for blocks is given by

$$\text{F-ratio} = \frac{\frac{BSS}{df_B}}{\frac{ESS}{df_e}} = \frac{\text{MBSS}}{\text{MESS}}$$

Mean Block Sum of Squares (s^2_b)

Mean Error Sum of Squares (s^2_e)

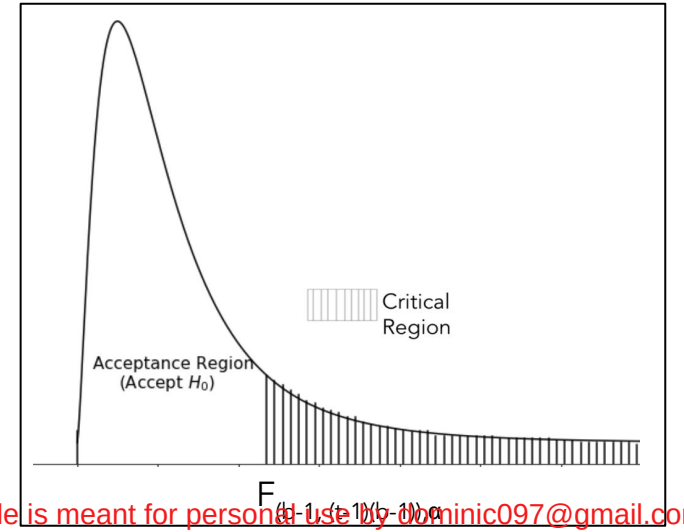
- Under H_0, the test statistic follows F-distribution with (df_B, df_e) degrees of freedom

- Decision Rule: If $F_{cal} \geq F_{(b-1,(t-1)(b-1))\alpha}$ then we reject H_0 at α% level of significance

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

Decision Rule: If $F_{cal} \geq F_{(b-1,\ (t-1)(b-1)),\alpha}$ or p-value ≤ α, then we reject H_0 at α% level of significance.



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

The two ANOVA table is given as follows:

Source of variation	Degrees of freedom	Sum of Squares	Mean Sum of Squares	F-ratio
Treatment	t-1	TrSS	s^2_t	$\frac{s_t^2}{s_e^2}$
Block	b-1	BSS	s^2_b	$\frac{s_b^2}{s_e^2}$
Error	(t-1)(b-1)	ESS	s^2_e	
Total	bt-1	TSS		

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA - procedure

1. State the hypothesis to be tested

2. Compute the sum of squares

   a. The total sum of squares, TSS = $\sum_{j=1}^{t}\sum_{i=1}^{b} (x_{ij} - \bar{x}_{..})^2$

   b. The treatment sum of squares TrSS = $\sum_{j=1}^{t} b\,(\bar{x}_{i.} - \bar{x}_{..})^2$

   c. The block sum of squares BSS = $\sum_{i=1}^{b} t\,(\bar{x}_{.j} - \bar{x}_{..})^2$

   d. The Error sum of squares, ESS = TSS - TrSS - BSS

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA - procedure

3. Compute mean sum of squares

   a. s^2_t = Mean treatment sum of squares (MTrSS) = TrSS/(t-1)

   b. s^2_b = Mean block sum of squares (MBSS) = BSS/(b-1)

   c. s^2_e = Mean error sum of squares (MESS) = ESS/[(t-1)(b-1)]

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA - procedure

4. Compute the F-ratios for

   a. Treatment: $\text{F-ratio}_{\text{Tr}} = \dfrac{\text{MTrSS}}{\text{MESS}} = \dfrac{s_t^2}{s_e^2}$

   b. Block: $\text{F-ratio}_{\text{B}} = \dfrac{\text{MBSS}}{\text{MESS}} = \dfrac{s_b^2}{s_e^2}$

5. Prepare the ANOVA table

6. Write the decision and conclusion accordingly

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

Question:

The business is thriving, and Ryan finds it crucial that the production should progress without any breaks. He decides to have three shifts - Morning, Afternoon and Night shift.

Ryan collects data of tensile strength (in N/m^2) from all the 4 machines and the 3 shifts as given.

Test at 5% level of significance.

	A	B	C	D
Morning Shift	68.7	78.7	55.9	84.7
Afternoon Shift	75.4	75.4	56.1	85.3
Night Shift	80.9	80.9	57.3	87.9

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

Solution:

There are 4 machines - A, B, C and D

Let α_1 be the average tensile strength due to machine A
α_2 be the average tensile strength due to machine B
α_3 be the average tensile strength due to machine C
α_4 be the average tensile strength due to machine D

To test, H_0: $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4$ Against H_1: At least one α_i is different (i=1, 2, 3, 4)

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

Solution:

There are 3 shifts - Morning, Afternoon and Night

Let β_1 be the average tensile strength due to Morning shift
β_2 be the average tensile strength due to Afternoon shift
β_3 be the average tensile strength due to Night shift

To test, H_0: $\beta_1 = \beta_2 = \beta_3$ Against H_1: At least one β_i is different (i=1, 2, 3)

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

Solution:

The grand mean:

$$\bar{x}_{..} = \frac{68.7+78.7+...+57.3+87.9}{12} = 73.93$$

The total sum of squares

$$TSS = \sum_i^t \sum_j^b (\bar{x}_{ij} - \bar{x}_{..})^2$$

$$= (68.7 - 73.93)^2 + \ldots + (87.9 - 73.93)^2$$

$$= 1511.567 \ (\mathrm{N/m^2})^2$$

	A	B	C	D
Morning Shift	68.7	78.7	55.9	84.7
Afternoon Shift	75.4	75.4	56.1	85.3
Night Shift	80.9	80.9	57.3	87.9

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

Solution:

The treatment sum of squares

$$TrSS = \sum_i^t b(\bar{x}_{i.} - \bar{x}_{..})^2$$

$$= 3(75 - 73.93)^2 + \ldots + 3(85.97 - 73.93)^2$$

$$= 1414.647 \ (\mathrm{N/m}^2)^2$$

	A	B	C	D
Morning Shift	68.7	78.7	55.9	84.7
Afternoon Shift	75.4	75.4	56.1	85.3
Night Shift	80.9	80.9	57.3	87.9
$\bar{x}_{i.}$	75	78.33	56.43	85.97

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

Solution:

The block sum of squares

	A	B	C	D	$\bar{x}_{.j}$
Morning Shift	68.7	78.7	55.9	84.7	72
Afternoon Shift	75.4	75.4	56.1	85.3	73.05
Night Shift	80.9	80.9	57.3	87.9	76.75

$$BSS = \sum_j^b t(\bar{x}_{.j} - \bar{x}_{..})^2$$

$$= 4(72 - 73.93)^2 + 4(73.05 - 73.93)^2 + 4(76.75 - 73.93)^2$$

$$= 49.807 \ (\mathrm{N/m}^2)^2$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

Solution:

The treatment sum of squares is

$$ESS = \sum_i^t \sum_j^b (x_{ij} - \bar{x}_{i.} - \bar{x}_{.j} + \bar{x}_{..})^2$$

$$= (68.7 - 75 - 72 - 68.365 + 73.93)^2 + \ldots$$
$$\ldots + (68.7 - 75 - 72 - 68.365 + 73.93)^2$$

$$= 47.1133 \ (\mathrm{N/m}^2)^2$$

	A	B	C	D	$\bar{x}_{.j}$
Morning Shift	68.7	78.7	55.9	84.7	72
Afternoon Shift	75.4	75.4	56.1	85.3	73.05
Night Shift	80.9	80.9	57.3	87.9	76.75
$\bar{x}_{i.}$	75	78.33	56.43	85.97	

Or it can be calculated as $\mathrm{ESS} = \mathrm{TSS} - \mathrm{TrSS} - \mathrm{BSS} = 47.1133 \ (\mathrm{N/m}^2)^2$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

Solution:

Source of variation	Degrees of freedom	Sum of Squares	Mean Sum of Squares	F-ratio
Treatment	t-1 = 4-1 = 3	TrSS = 1414.647	$s_t^2 = \frac{1414.647}{3} = 471.5489$	$\frac{s_t^2}{s_e^2} = 60.0529$
Block	b-1 = 3-1 = 2	BSS = 49.8067	$s_b^2 = \frac{49.8067}{2} = 24.903$	$\frac{s_b^2}{s_e^2} = 3.1715$
Error	(b-1)(t-1) = 6	ESS = 47.1133	$s_e^2 = \frac{47.1133}{6} = 7.852$	
Total	bt-1 = 12 -1 =11	TSS = 1511.567		

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

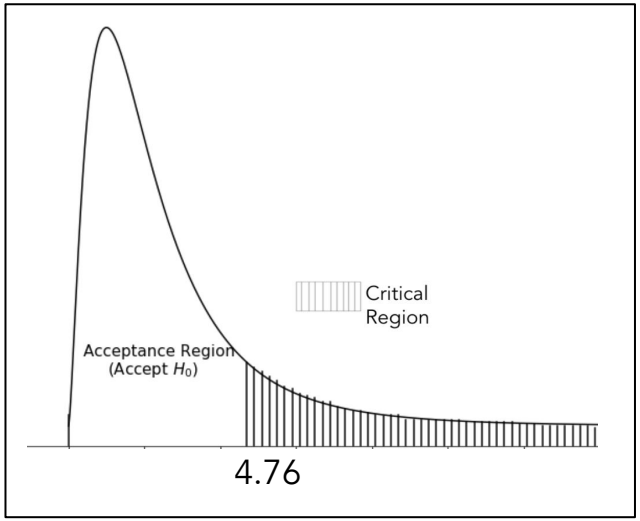
Solution:

For treatments, F-ratio = 60.0529

From the F table we have
$F_{(t-1,(b-1)(t-1)),\alpha} = F_{(3,6),0.05} = 4.76$

Since $F > F_{(3,6),\alpha}$ (60.05 > 4.76), we reject H_0

There is enough evidence to conclude that the tensile strength due to machines is different



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Two way ANOVA

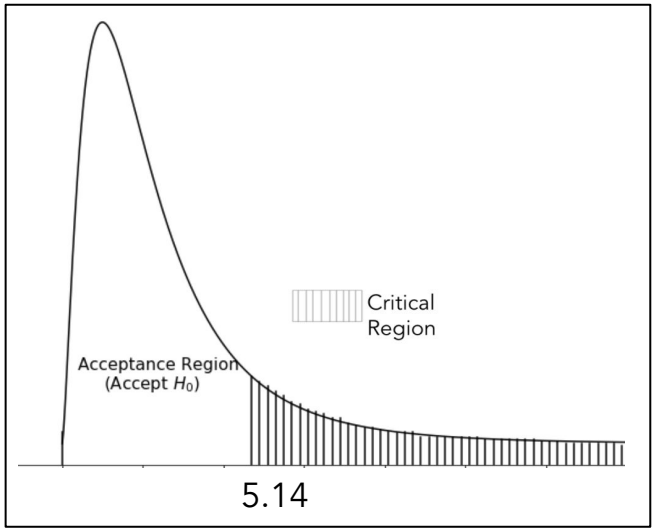
Solution:

For Blocks, F-ratio = 3.17

From the F table we have
$F_{(b-1,(b-1)(t-1)),\alpha} = F_{(2,6),0.05} = 5.14$

Since $F < F_{(2,6),\alpha}$ (3.17 < 5.14), we fail to reject H_0

There is no enough evidence to conclude that the tensile strength due to shifts is different



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Violations to assumptions in ANOVA

- Normality of data:
  When the group sizes are equal and sufficiently large, ANOVA is robust against normality the normality assumption. In this case, the samples should be similar in shape i.e. both positively or negatively skewed.

- Homogeneity of variances:
  When the group sizes are vastly unequal, the F statistic would be biased and will cause the null hypothesis to be falsely rejected.

If violation occurs, conducting a non-parametric equivalent of the analysis is more appropriate.

- Independence of samples:
  If this assumption is violated, we run into the risk of wrong results.

- Sample sizes:
  ANOVA is robust to moderate departures from this assumption.
  As long as the ratio between the largest sample size and the smallest sample size is less than 1.5, there shouldn't be issues.

- SS formula slightly changes, but the statistical tools automatically handle it.
- In place of using a normal mean, use a weighted mean.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

ANOVA *extensions*

ANCOVA:

- One or more independent variables are categorical
- The dependent variable is continuous.

Example:

For three groups that receive a drug, are the results significantly different ?

The question can be answered with ANOVA, but ANCOVA allows for additional controls, that might influence the outcome but have nothing to do with the drug, such as lifestyle healthiness.

MANOVA:

- **Multivariate analysis of variance (MANOVA)** is an extension of ANOVA.
- In an ANOVA, we examine for statistical differences on one continuous dependent variable by an independent grouping variable.
- The MANOVA extends this analysis by taking into account multiple continuous dependent variables, and bundles them together into a weighted linear combination or composite variable.

Example: In the previous example of college marks, adding a column of gender of the students

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

HAPPY LEARNING

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

SRM INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

Thank You

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.